

[1]にて: `import pandas as pd`
`import numpy as np`
`import matplotlib.pyplot as plt`

[2]にお... `df = pd.read_csv("data/ETTh1.csv")`

[3]では: `df.head()`

アウト[3...

	日付	HUFL	船体	MUFL	マル	LUFL	静かに	OT
0	2016-07-01 00:00:00	5.827	2.009	1.599	0.462	4.203	1.340	30.531000
1	2016-07-01 01:00:00	5.693	2.076	1.492	0.426	4.142	1.371	27.787001
2	2016-07-01 02:00:00	5.157	1.741	1.279	0.355	3.777	1.218	27.787001
3	2016-07-01 03:00:00	5.090	1.942	1.279	0.391	3.807	1.279	25.044001
4	2016-07-01 04:00:00	5.358	1.942	1.492	0.462	3.868	1.279	21.948000

[4]にて: `df.shape`

アウト[4... (17420, 8)

[5]にお... `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 17420 entries, 0 to 17419
Data columns (total 8 columns):
#   Column  Non-Null Count  Dtype
---  -
0    date    17420 non-null   str
1   HUFL    17420 non-null   float64
2   HULL    17420 non-null   float64
3   MUFL    17420 non-null   float64
4   MULL    17420 non-null   float64
5   LUFL    17420 non-null   float64
6   LULL    17420 non-null   float64
7    OT      17420 non-null   float64
dtypes: float64(7), str(1)
memory usage: 1.1 MB
```

[6]にお... `df.isnull().sum()`

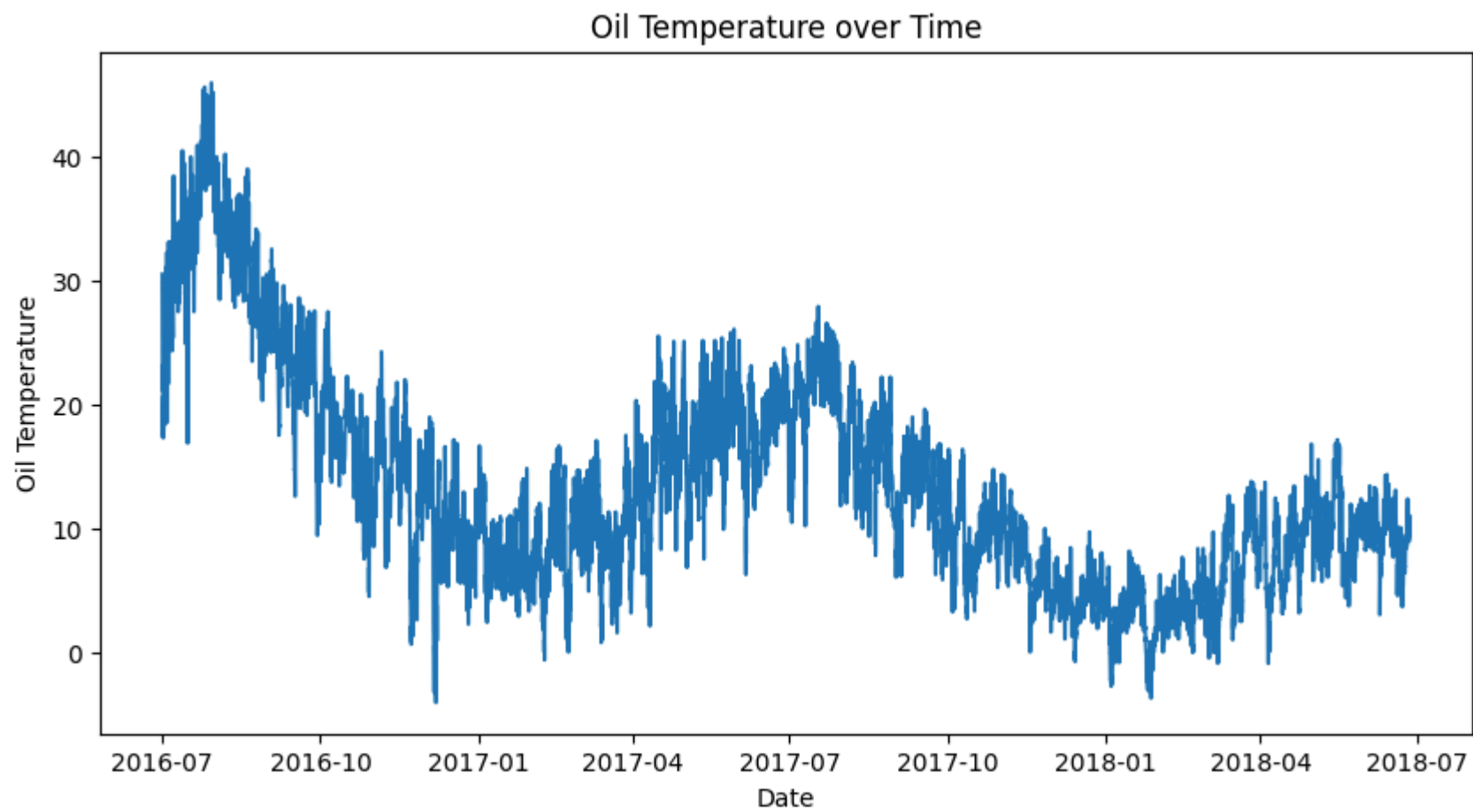
アウト[6...
date 0
HUFL 0
HULL 0
MUFL 0
MULL 0
LUFL 0
LULL 0
OT 0
dtype: int64

[7]にお... `df["date"] = pd.to_datetime(df["date"])`

[9]にて: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 17420 entries, 0 to 17419
Data columns (total 8 columns):
 #   Column  Non-Null Count  Dtype  
---  -
 0   date    17420 non-null   datetime64[us]
 1   HUFL    17420 non-null   float64
 2   HULL    17420 non-null   float64
 3   MUFL    17420 non-null   float64
 4   MULL    17420 non-null   float64
 5   LUFL    17420 non-null   float64
 6   LULL    17420 non-null   float64
 7   OT      17420 non-null   float64
dtypes: datetime64[us](1), float64(7)
memory usage: 1.1 MB
```

```
[16]にて: plt.figure(figsize=(10, 5))
plt.plot(df["date"], df["OT"])
plt.xlabel("Date")
plt.ylabel("Oil Temperature")
plt.title("Oil Temperature over Time")
plt.show()
```



```
In [17]: df.describe()
```

Out[17]:

	date	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
count	17420	17420.000000	17420.000000	17420.000000	17420.000000	17420.000000	17420.000000	17420.000000
mean	2017-06-28 21:30:00	7.375141	2.242242	4.300239	0.881568	3.066062	0.856932	13.324672
min	2016-07-01 00:00:00	-22.705999	-4.756000	-25.087999	-5.934000	-1.188000	-1.371000	-4.080000
25%	2016-12-29 10:45:00	5.827000	0.737000	3.296000	-0.284000	2.315000	0.670000	6.964000
50%	2017-06-28 21:30:00	8.774000	2.210000	5.970000	0.959000	2.833000	0.975000	11.396000
75%	2017-12-27 08:15:00	11.788000	3.684000	8.635000	2.203000	3.625000	1.218000	18.079000
max	2018-06-26 19:00:00	23.643999	10.114000	17.341000	7.747000	8.498000	3.046000	46.007000
std	NaN	7.067744	2.042342	6.826978	1.809293	1.164506	0.599552	8.566946

In [18]: `df.corr(numeric_only=True)`

Out[18]:

	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
HUFL	1.000000	0.114672	0.987355	0.066002	0.291418	0.023606	0.059916
HULL	0.114672	1.000000	0.068817	0.930491	0.259487	0.377641	0.224354
MUFL	0.987355	0.068817	1.000000	0.046266	0.177491	-0.046519	0.050854
MULL	0.066002	0.930491	0.046266	1.000000	0.128607	0.069419	0.220004
LUFL	0.291418	0.259487	0.177491	0.128607	1.000000	0.334563	0.118836
LULL	0.023606	0.377641	-0.046519	0.069419	0.334563	1.000000	0.067455
OT	0.059916	0.224354	0.050854	0.220004	0.118836	0.067455	1.000000

In [19]: `correlation = df.corr(numeric_only=True)`
`correlation["OT"].sort_values(ascending=False)`

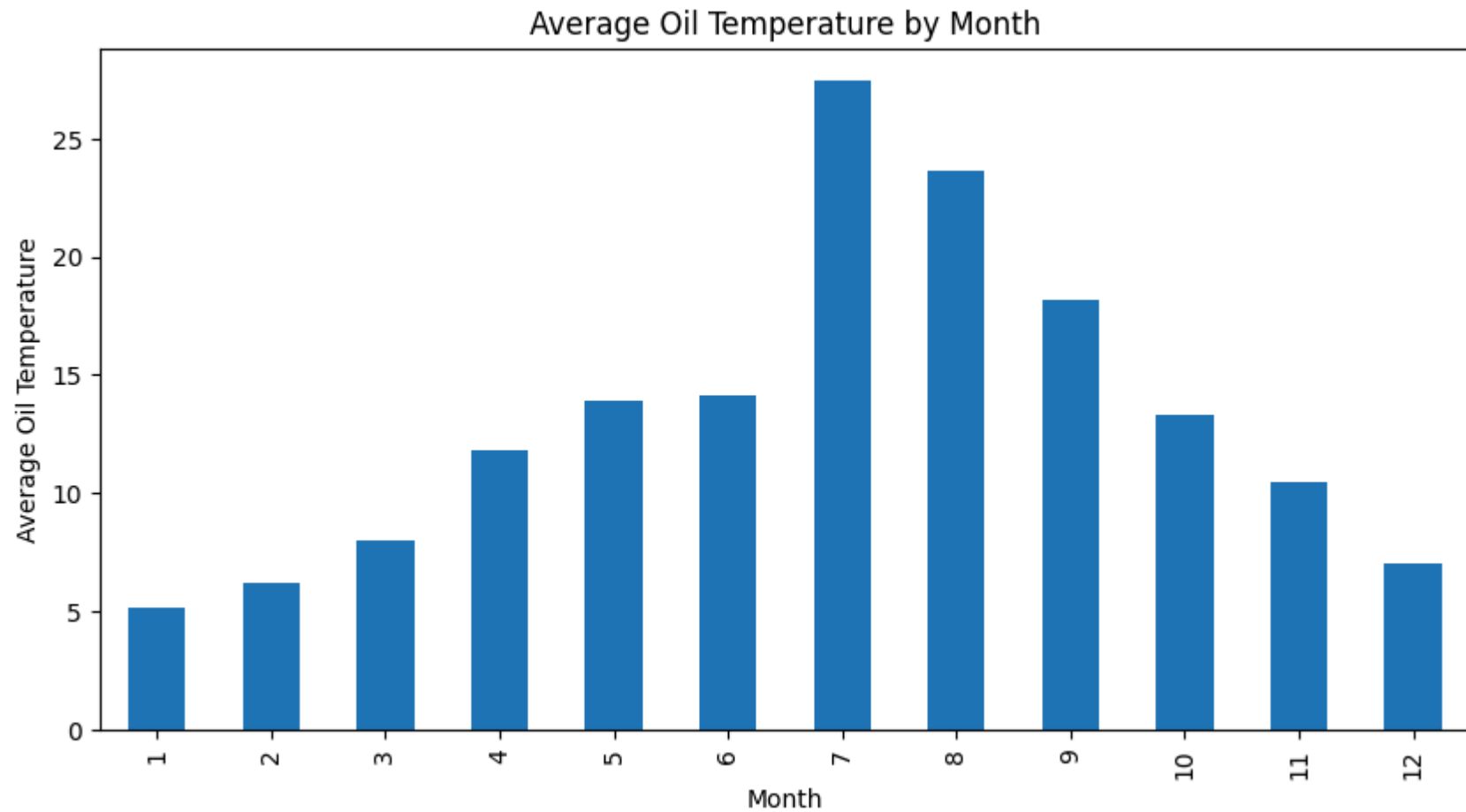
```
Out[19]: OT      1.000000
        HULL    0.224354
        MULL    0.220004
        LUFL    0.118836
        LULL    0.067455
        HUFL    0.059916
        MUFL    0.050854
        Name: OT, dtype: float64
```

```
In [20]: df["hour"] = df["date"].dt.hour
        df["dayofweek"] = df["date"].dt.dayofweek
        df["month"] = df["date"].dt.month
```

```
In [21]: monthly_ot = df.groupby("month")["OT"].mean()

        monthly_ot.plot(kind="bar", figsize=(10, 5))

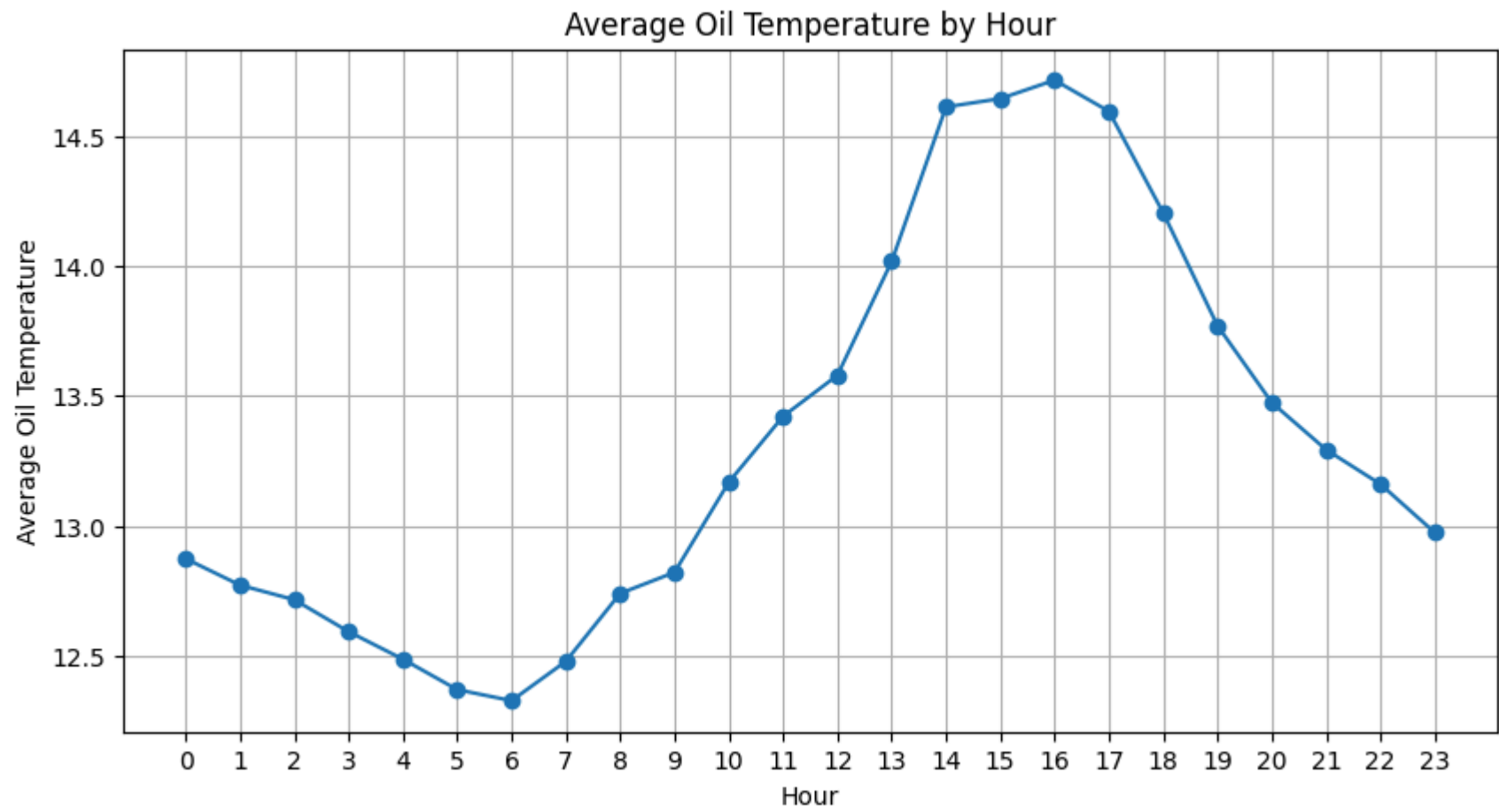
        plt.xlabel("Month")
        plt.ylabel("Average Oil Temperature")
        plt.title("Average Oil Temperature by Month")
        plt.show()
```



```
In [25]: hourly_ot = df.groupby("hour")["OT"].mean()

hourly_ot.plot(figsize=(10, 5), marker="o")

plt.xlabel("Hour")
plt.ylabel("Average Oil Temperature")
plt.title("Average Oil Temperature by Hour")
plt.xticks(range(24))
plt.grid()
plt.show()
```



In []: